

Maxglow Ecommerce Platform

Admin Panel - Technical Perspective Documentation

1. Admin Dashboard Architecture

Technology:

- React.js
- Next.js
- Bootstrap 5
- Redux Toolkit

Purpose:

Centralized business management system.

Admin can monitor:

- Total sales
- Revenue
- Orders
- Customers
- Inventory status
- Analytics

2. Admin Authentication & Authorization

Technology:

- JWT Authentication
- bcrypt Password Encryption
- Role Based Access Control

Roles:

Super Admin:

Full system access

Manager:

Orders, inventory, reports

Staff:

Limited operational access

3. Admin Frontend Flow

Admin Action

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React Component

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Redux State

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API Request

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Backend Validation

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Database Update

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UI Refresh

4. Backend Architecture

Technology:

- Node.js
- Express.js

Responsibilities:

- Admin APIs
- Business logic
- Data processing
- Security handling

Flow:

Admin Panel

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Express API

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Controller

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Service Layer

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MongoDB

5. Product Management System

Admin Features:

- Add product
- Edit product
- Delete product

- Update pricing
- Manage categories
- Upload images/videos

Product data stored in MongoDB.

6. Inventory Management System

Features:

- Stock tracking
- Stock update
- Low stock alert
- Batch management
- Expiry tracking

Flow:

Order Created

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Inventory Update

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Stock Quantity Reduced

7. Order Management System

Admin can:

- View orders
- Confirm orders
- Update delivery status
- Cancel orders
- Manage refunds

Order Flow:

Placed

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Confirmed

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Packed

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Shipped

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Delivered

8. Customer Management

Admin can view:

- Customer profiles
- Order history
- Purchase activity
- Address information

9. Payment Management

Technology:

Razorpay Integration

Admin can monitor:

- Successful payments
- Failed payments
- Transaction status
- Refund details

10. Reports & Analytics

Features:

- Sales reports
- Product performance
- Customer analytics
- Revenue analysis

Export support:

PDF / Excel

11. Media Storage System

Technology:

AWS S3

Used for:

- Product images
- Product videos
- Documents

Flow:

Upload File

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AWS S3 Storage

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Save URL in MongoDB

12. Security Implementation

Features:

- JWT protected routes
- API validation
- Helmet security
- Rate limiting
- Activity logs

13. Database Architecture

MongoDB Collections:

Users

Products

Orders

Inventory

Payments

Reviews

Coupons

Logs

14. Deployment Architecture

Frontend:

Next.js + Vercel

Backend:

Node.js + AWS

Database:

MongoDB Atlas

Storage:

AWS S3

15. Performance Optimization

Implementation:

- API caching
- Pagination
- Database indexing
- Image optimization
- Lazy loading

16. Scalability

Future support:

- Multiple warehouses
- Mobile application
- Advanced analytics
- AI based business insights

Final Admin Architecture:

Admin User

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Next.js Admin Panel

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Node.js Express API

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MongoDB

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AWS S3

Production ready ecommerce management architecture.